



Admas University
UNIVERSITY FACULTY OF INFORMATICS
DEPARTMENT OF COMPUTER SCIENCE

Introduction to Machine Learning

Lecture 1

Introduction to Machine Learning

October, 2025



What is Learning?

"The activity or process of gaining knowledge or skill by studying, practicing, being taught, or experiencing something." – Merriam Webster dictionary

"Learning is any process by which a system improves performance from experience." – Herbert Simon

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ." – Tom Mitchell

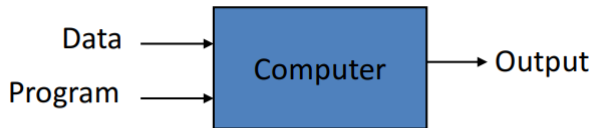
What is Machine Learning

- Machine learning is a subset of [Artificial Intelligence \(AI\)](#) which provides machines the ability to learn automatically & improve from experience [without being explicitly programmed](#) to do so.
- In the sense, it is the practice of getting Machines to solve problems by gaining the ability to think.
- If you feed a machine a good amount of data, it will learn how to interpret, process and analyze this data by using [Machine Learning Algorithms](#), in order to solve real-world problems.

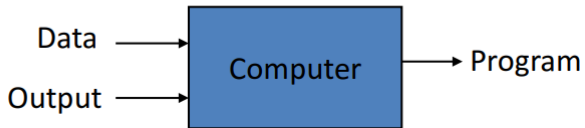
What is Machine Learning

- For many problems, it's difficult to program the correct behavior by hand
 - recognizing people and objects
 - understanding human speech
- Machine learning approach: program an algorithm to automatically learn from data, or from experience
- Why might you want to use a learning algorithm?
 - hard to code up a solution by hand (e.g. vision, speech)
 - system needs to adapt to a changing environment (e.g. spam detection)
 - want the system to perform better than the human programmers
 - privacy/fairness (e.g. ranking search results)

Traditional Programming



Machine Learning



What is Machine Learning

- It's similar to statistics...
 - Both fields try to uncover patterns in data
 - Both fields draw heavily on calculus, probability, and linear algebra, and share many of the same core algorithms
- But it's not statistics!
 - Stats is more concerned with helping scientists and policymakers draw good conclusions; ML is more concerned with building autonomous agents
 - Stats puts more emphasis on interpretability and mathematical rigor; ML puts more emphasis on predictive performance, scalability, and autonomy

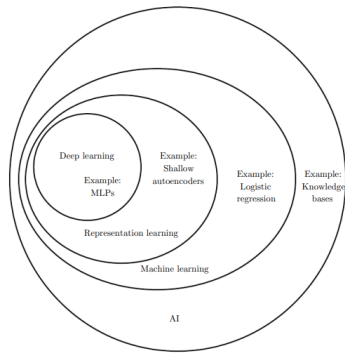
Relations to AI

Nowadays, “machine learning” is often brought up with “artificial intelligence” (AI)

AI does not always imply a learning based system

- Symbolic reasoning
- Rule-based system
- Tree search etc

Learning based system → learned based on the data
→ more flexibility, good at solving pattern recognition problems



Relations to human learning

- Human learning is:
 - Very data efficient
 - An entire multitasking system (vision, language, motor control, etc.)
 - Takes at least a few years
- For serving specific purposes, machine learning doesn't have to look like human learning in the end.
- It may borrow ideas from biological systems, e.g., neural networks.
- It may perform better or worse than humans.

Machine Learning Terminologies

- **Algorithm:** A Machine Learning algorithm is a **set of rules** and **statistical techniques** used to learn patterns from data and draw significant information from it. It is the logic behind a Machine Learning model.
- **Model:** A model is the main component of Machine Learning. A model is trained by using a **Machine Learning Algorithm**. An algorithm **maps all the decisions that a model is supposed to take based on the given input**, in order to get the correct output.

Machine Learning Terminologies

- **Predictor Variable**: It is a feature(s) of the data that can be used to predict the output.
- **Response Variable**: It is the feature or the output variable that needs to be predicted by using the predictor variable(s).
- **Training Data**: The Machine Learning model is built using the training data. The training data helps the model to identify key trends and patterns essential to predict the output.
- **Testing Data**: After the model is trained, it must be tested to evaluate how accurately it can predict an outcome. This is done by the testing data set.
- **Validation data**: is a subset of the training data used to tune the model's **hyperparameters** and evaluate the model's performance during the training process. The validation data helps prevent **overfitting** by providing an independent measure of how well the model is generalizing to unseen data.

History of Machine Learning

- 1940s and 1950s ML was known as **cybernetics**:
 - 1943 – McCulloch-Pitts Neuron: a linear model that recognizes two categories as +ve or -ve.
- 1957 – Perceptron algorithm (implemented as a circuit!)
 - Developed by Frank Rosenblatt, an early model of a single-layer neural network.
 - It was designed to mimic how a biological neuron works
- 1959 – Arthur Samuel wrote a learning-based checkers program that could defeat him
 - The machine learns from experience
 - used an algorithm called **minimax search**
- 1969 – Minsky and Papert's book Perceptrons (limitations of linear models)
 - Minsky and Papert highlighted a crucial limitation of the single-layer perceptron (solve only linearly separable problems)
 - Ushered the **AI Winter** but had more nuanced impact on the field.

History of Machine Learning

- 1980s – Some foundational ideas
 - **Connectionist** psychologists explored neural models of cognition
 - Arose in teh context of **cognitive science**
 - **Main Idea**: a large number of simple computational units can achieve intelligent behaviour when networked together.
 - The successful use of **back-propagation** to train DNNs with internal representation and the populariztion of **BP algorithm** (by Geoffrey Hinton and colleagues in 1988)
 - 1984 – Leslie Valiant formalized the problem of learning as Probably Approximately Correct (PAC) learning
 - introduced key concepts like "probably" (with a high probability) and "approximately" (with a small error)
 - 1988 – Judea Pearl's book Probabilistic Reasoning in Intelligent Systems introduced Bayesian networks
 - powerful graphical model for representing and reasoning with probabilistic relationships between variables

History of machine learning

- 1990s – important advances in modeling **sequences** with NNs
 - In 1997 LSTM was introduced.
 - Second wave of NNs research lasted until mid-1990s (another **AI winter**).
 - NN-based ventures began to make unrealistically ambitious claims while seeking investments.
 - **Lack of data** and collapse of the specialized **AI hardware** market were also major contributors
- 2000s – The third wave of AI and NN research characterized by a focus on deep, complex architectures, driven by three key factors: **the availability of Big Data, the rise of powerful computing, and significant algorithmic improvements**.
 - The use of the term *deep learning* was popularized
 - **ImageNet Moment (2012)**: **AlexNet**, a deep CNN, won the ImageNet competition, drastically lowering the error rate and proving deep learning's superiority in image classification.
 - What followed was equivalent to the cambrian evolution in the AI ecosystem

History of machine learning

- Generative AI and The Transformer Era (2017 – Present):
 - The Transformer Architecture (2017): A new model architecture that proved highly effective for sequential data.
 - Large Language Models (LLMs): The Transformer architecture led to the development of models like GPT (2018), which could generate human-like text at unprecedented scale and coherence.
 - Diffusion Models (2020-Present): A new class of generative models that work by gradually "denoising" an image. Models like DALL-E 2 and Stable Diffusion (2022) use this technique to generate stunningly creative images from text prompts.

Essential math for machine learning

- **Linear algebra:** Linear algebra deals with mathematical operations on vectors and matrices. It is used in machine learning for tasks such as matrix operations, vector operations, and solving systems of linear equations.
- **Calculus:** Calculus deals with the study of how things change. It includes concepts such as derivatives (which measure the rate of change of a function) and integrals (which measure the area under a curve). Calculus is used in machine learning for tasks such as optimization and calculating gradients.
- **Probability theory:** Probability theory deals with the study of randomness and uncertainty. It includes concepts such as probability distributions (which describe the likelihood of different outcomes), and conditional probability

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

- Probability theory is used in machine learning for tasks such as statistical inference, generative models, and decision making.

Essential math for machine learning

- **Statistics:** Statistics deals with the collection, analysis, and interpretation of data. It includes concepts such as descriptive statistics (which summarize and describe data), hypothesis testing (which is used to make inferences about populations based on sample data), and regression analysis (which is used to model the relationship between variables). Statistics is used in machine learning for tasks such as data analysis, statistical inference, and model evaluation.
- **Information theory:** Information theory deals with the quantification of data. It includes concepts such as entropy (which measures the uncertainty of a random variable), mutual information (which measures the amount of information shared by two random variables), and the Kullback-Leibler divergence (which measures the difference between two probability distributions). Information theory is used in machine learning for tasks such as feature selection, data compression, and data encoding.

Essential math for machine learning

- **Optimization:** Optimization is the process of finding the best solution to a problem. It includes concepts such as gradient descent (which is used to find the minimum of a function) and stochastic gradient descent (which is a variation of gradient descent that uses a random subset of data to estimate the gradient). Optimization is used in machine learning for tasks such as finding the best model parameters and minimizing the loss function.
- **Graph theory:** Graph theory deals with the study of networks and connectivity. It includes concepts such as graphs (which are mathematical structures that model relationships between objects), nodes (which are the objects in a graph), edges (which are the connections between objects in a graph), and centrality (which measures the importance of nodes in a graph). Graph theory is used in machine learning for tasks such as clustering, community detection, and network analysis.

Applications of machine learning

- Machine learning has a wide range of applications across different industries and fields. Here are some examples:
 - **Image and Speech Recognition:** Machine learning is used to recognize images and speech in real-time. It is used in applications such as facial recognition, voice recognition, and object detection.
 - **Fraud Detection:** Machine learning is used to detect fraud in financial transactions. It can be used to identify fraudulent transactions, detect unusual patterns, and flag suspicious activities.
 - **Healthcare:** Machine learning is used in healthcare for diagnosis, treatment planning, and drug development. It can be used to analyze medical images, predict patient outcomes, and develop personalized treatment plans.
 - **Predictive Maintenance:** Machine learning is used in industrial settings to predict when machines will fail and need maintenance. This helps reduce downtime and improve productivity.

Applications of machine learning

- **Natural Language Processing:** Machine learning is used in natural language processing to enable machines to understand human language. It is used in applications such as chatbots, voice assistants, and language translation.
- **Marketing:** Machine learning is used in marketing to analyze customer behavior and preferences. It can be used to personalize marketing campaigns and recommend products to customers.
- **Autonomous Vehicles:** Machine learning is used in autonomous vehicles to enable them to perceive their environment and make decisions. It is used in applications such as self-driving cars and drones.
- **Financial Forecasting:** Machine learning is used in finance to forecast stock prices, predict market trends, and identify investment opportunities.

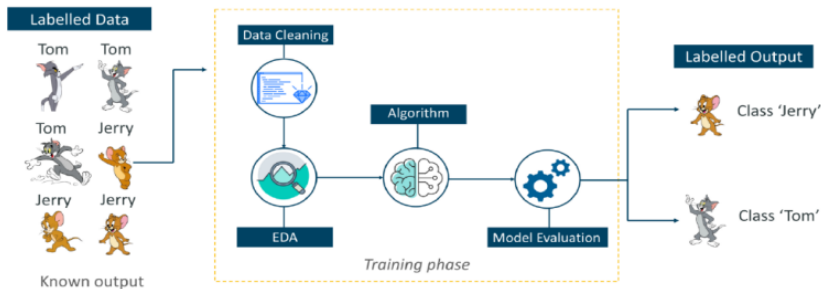
Types of machine learning

- A machine can learn to solve a problem by following any one of the following four techniques.
- These are three ways in which a machine can learn:
 1. Supervised Learning
 2. Unsupervised Learning
 3. Semi-supervised learning
 4. Reinforcement Learning

Supervised Learning

- Supervised learning is a technique in which we teach or train the machine using data which is well **labeled**.
- To understand Supervised Learning let's consider an example.
- As kids need guidance to solve math problems. The teachers helped them understand what addition is and how it is done. Similarly, you can think of supervised learning as a type of Machine Learning that involves a guide.
- Types of problem
 - Regression
 - Classification

Supervised Learning



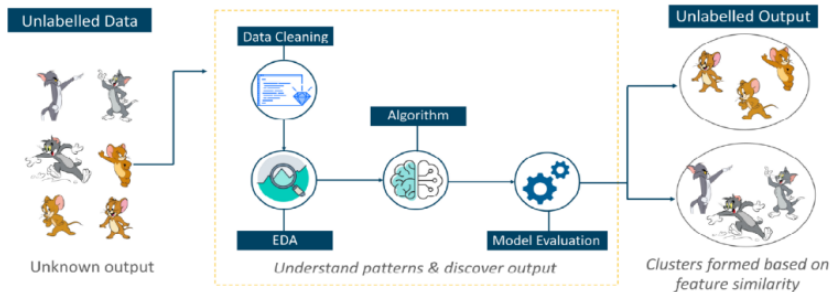
Supervised Learning

Here we're feeding the machine images of Tom and Jerry and the goal is for the machine to identify and classify the images into two groups (Tom images and Jerry images). The training data set that is fed to the model is labeled, as in, we're telling the machine, 'this is how Tom looks and this is Jerry'. By doing so you're training the machine by using labeled data.

Unsupervised Learning

- Unsupervised learning involves training by using unlabeled data and allowing the model to act on that information without guidance.
- Think of unsupervised learning as a smart kid that learns without any guidance.
- In this type of Machine Learning, the model is not fed with labeled data, as in the model has no clue that 'this image is Tom and this is Jerry', it figures out patterns and the differences between Tom and Jerry on its own by taking intentions of data.
- Types of Problem
 - Clustering
 - Association

Unsupervised Learning



Unsupervised Learning

For example, it identifies prominent features of Tom such as pointy ears, bigger size, etc, to understand that this image is of type 1. Similarly, it finds such features in Jerry and knows that this image is of type 2. Therefore, it classifies the images into two different classes without knowing who Tom is or Jerry is.

Semi-supervised Learning

- Semi-supervised learning offers a happy medium between supervised and unsupervised learning.
- During training, it uses a smaller labeled data set to guide classification and feature extraction from a larger, unlabeled data set.
- Semi-supervised learning can solve the problem of having not enough labeled data (or not being able to afford to label enough data) to train a supervised learning algorithm.
- Types of problem
 - Combination of tasks in both supervised and unsupervised

Reinforcement Learning

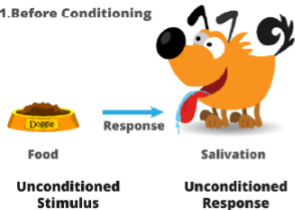
- Reinforcement learning can be thought of as a try and error method of learning. The machine gets a Reward or Penalty point for each action it performs. If the option is correct, the machine gains the reward point or gets a penalty point in case of a wrong response.
- The reinforcement learning algorithm is all about the interaction between the environment and the learning agent. The learning agent is based on **exploration and exploitation**.
- Exploration is when the learning agent acts on trial and error and Exploitation is when it performs an action based on the knowledge gained from the environment.
- The environment rewards the agent for every correct action, which is the reinforcement signal. With the aim of collecting more rewards obtained, the agent improves its environment knowledge to choose or perform the next action.

Reinforcement Learning Example

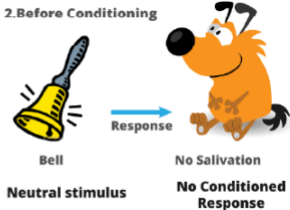
- **Let's see how Pavlov trained his dog using reinforcement training?**
- Pavlov divided the training of his dog into four stages.
 1. In the first part, Pavlov gave meat to the dog, and in response to the meat, the dog started salivating.
 2. In the next stage he created a sound with a bell, but this time the dogs did not respond anything.
 3. In the third stage, he tried to train his dog by using the bell and then giving them food. Seeing the food the dog started salivating.
 4. Eventually, the dogs started salivating just after hearing the bell, even if the food was not given as the dog was reinforced that whenever the master will ring the bell, he will get the food.

Reinforcement Learning Example

1. Before Conditioning



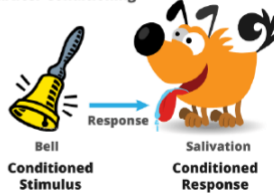
2. Before Conditioning



3. During Conditioning



4. After Conditioning



Assignment!

- Install the following:
 - Anaconda
 - `scikit-learn`, `numpy`, `pandas`, `matplotlib`, `jupyter`



Thank You

